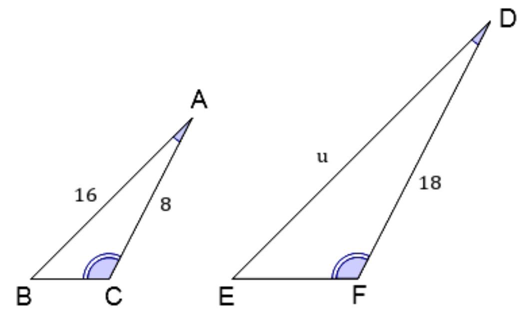
Applications of similar triangles

1 Find the value of u using a proportion statement.

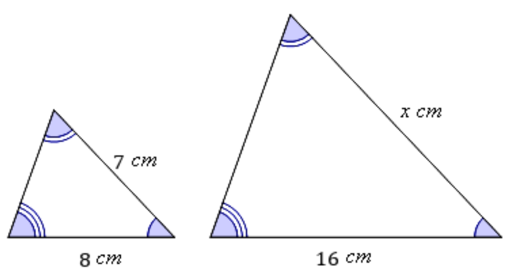


2 Consider the given similar triangles.

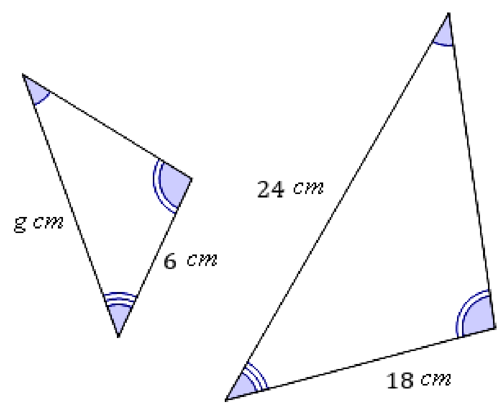
i Find the enlargement factor applied to each side of the smaller triangle.

ii Solve for the value of the variable.

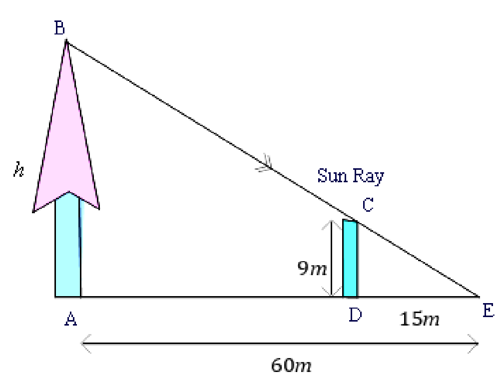
a



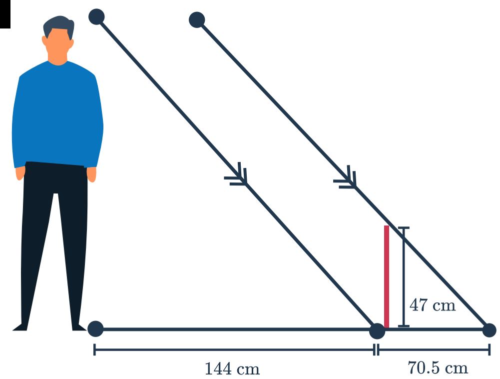
b



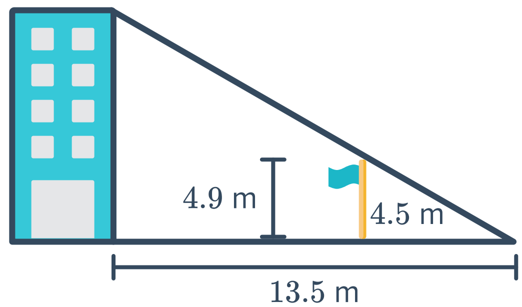
3 A large tree casts a shadow of 60 m. At the same time, a 9 m high fence, standing vertically, casts a shadow of 15 m.
Find h, the height of the tree.



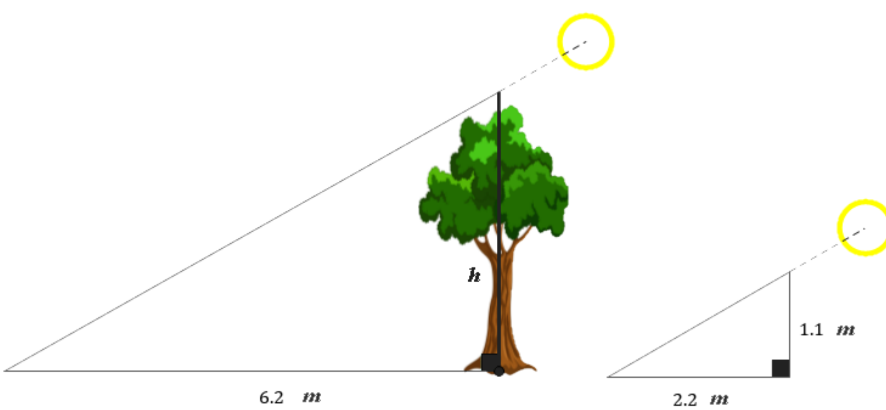
4 A man wants to measure his height. He stands in the sun and takes note of where his shadow ends and places a stick vertically in the ground at this point. He then measures the height and length of the stick and its shadow. The figure shows the results.
Find h, the height of the man.



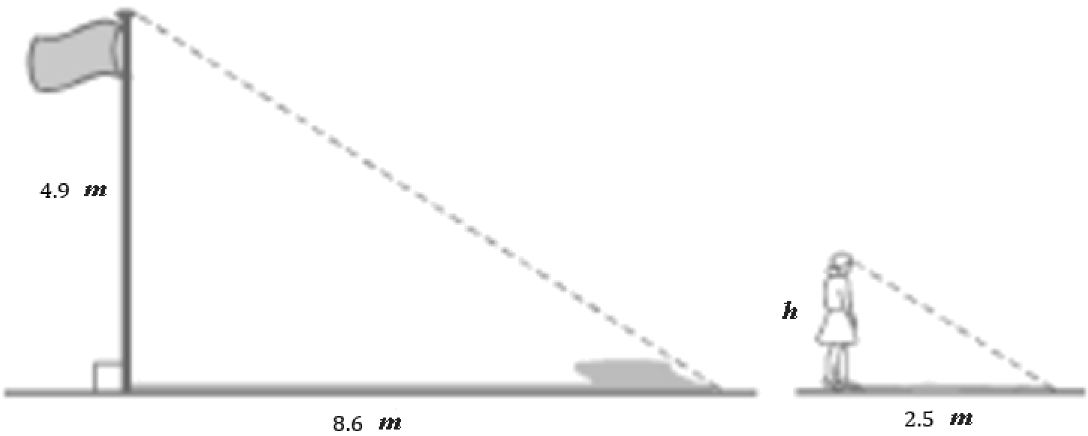
5 A $4.9\ \mathrm{m}$ high flagpole casts a shadow of $4.5\ \mathrm{m}$. At the same time, the shadow of a nearby building falls at the same point (S). The shadow cast by the building measures $13.5\ \mathrm{m}$.
Find h, the height of the building, using a proportion statement.



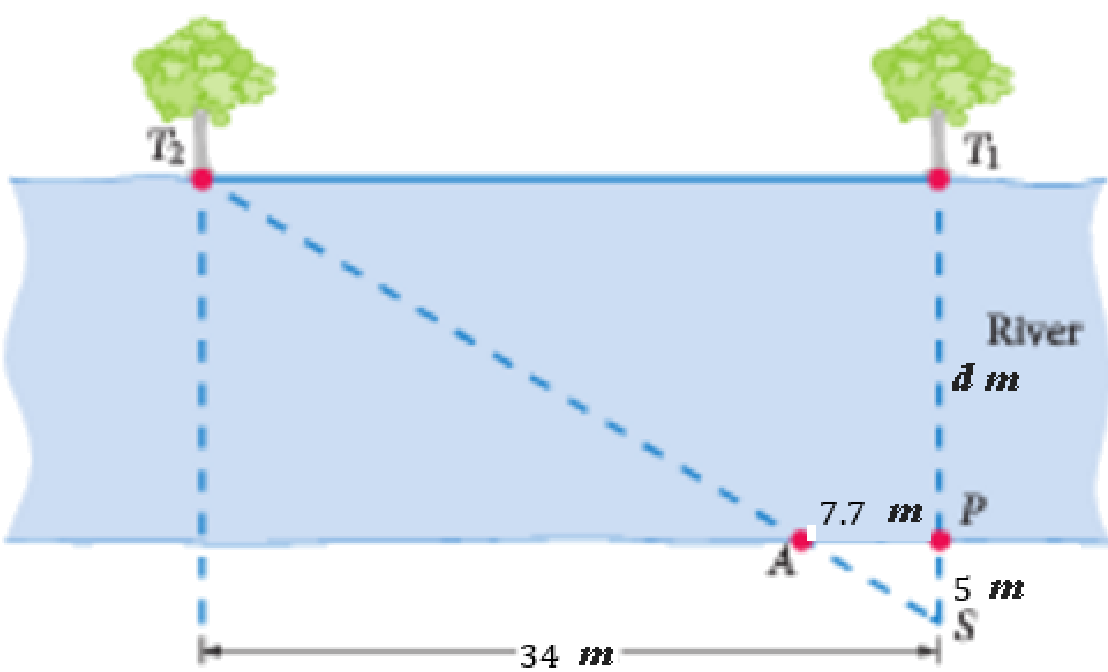
6 A stick of height $1.1\ \mathrm{m}$ casts a shadow of length $2.2\ \mathrm{m}$. At the same time, a tree casts a shadow of $6.2\ \mathrm{m}$.
Find the height, h of the tree.



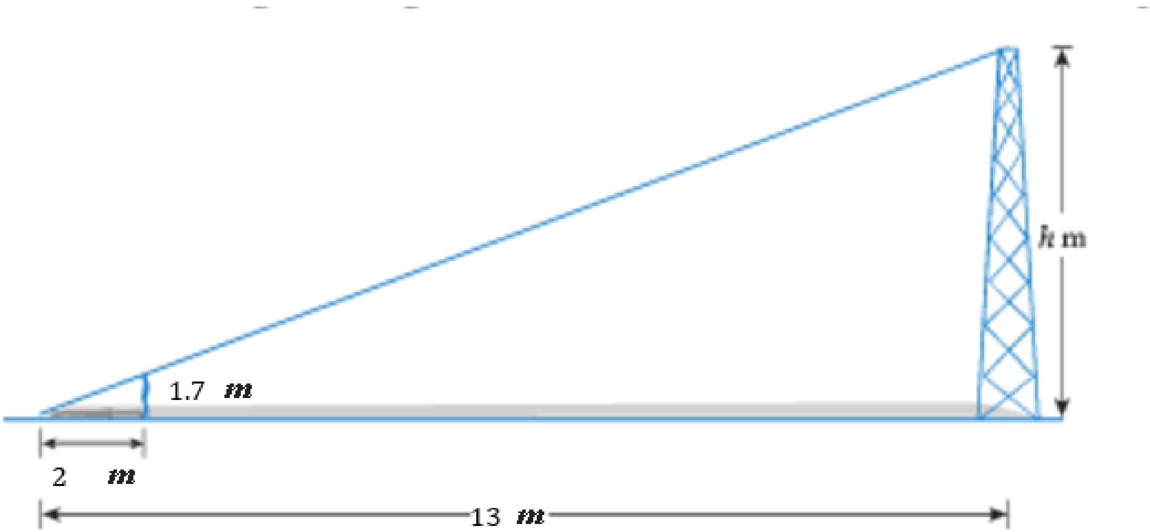
7 A 4.9 m flagpole casts a shadow of 8.6 m. Ar[redacted] casts a shadow of 2.5 m.
If Amelia is h meters tall, solve for h, correct to one decimal place.



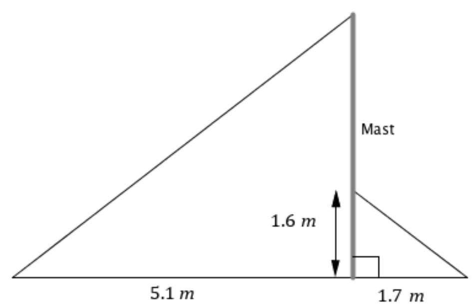
8 A surveyor needs to measure the distance across a river. There are two trees on the opposite bank that are 34 m apart. She stands 5 m from the bank, directly opposite the first tree. Her assistant has to move 7.7 m along the bank to place a stick directly in her line of sight to the second tree.
Find the width d of the river, correct to the nearest meter.



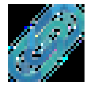
9 James is 1.7 m tall and casts a shadow 2 m long. At the same time, a tower casts a shadow 13 m long.
If the tower is h meters high, solve for h, correct to one decimal place.

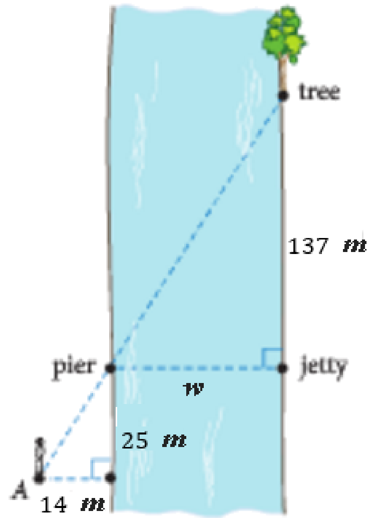


10 Two similar triangles are created by cables supporting a yacht's mast.
Solve for h, the height of the mast.

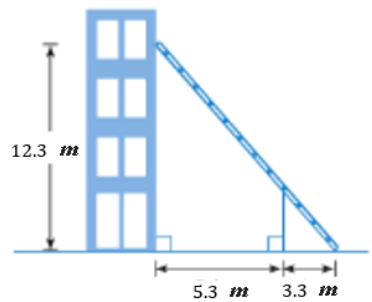


11 James stands at point A, as shown in the adjacent figure, so that he is in line with the pier and the tree on the other side of the canal. He then measures the direct distance to the edge of the water to be $14 \mathrm{~m}$, and the distance along the canal to be $25 \mathrm{~m}$ as shown. The distance of the tree from the jetty is known to be $137 \mathrm{~m}$.
If the width of the canal is w meters, solve for w. Round your answer to the nearest meter.





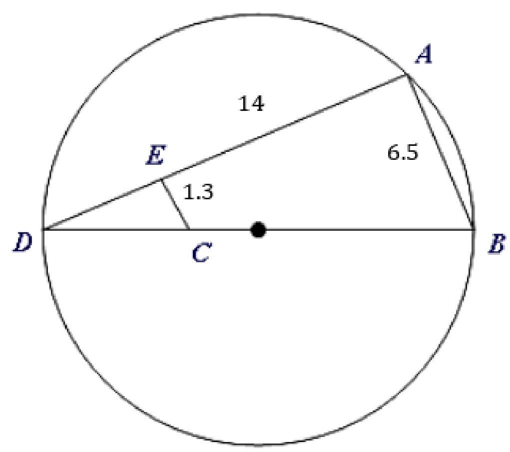
12 A fireman's ladder reaches $12.3 \mathrm{~m}$ up a building. The ladder is propped up by a support, placed 5.3 m from the building and $3.3 \mathrm{~m}$ from the base of the ladder.
If the support is L meters long, solve for L. Round your answer to one decimal place.



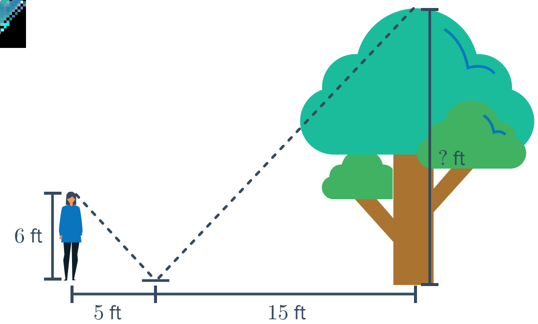
13 In the diagram, $\triangle ABD$ and $\triangle ECD$ are similar right-angled triangles, with $AE = 14$, $AB = 6.5$ and $EC = 1.3$.

a Let x be the length of $\overline{ED}$. Form an equation and find the exact length of x.

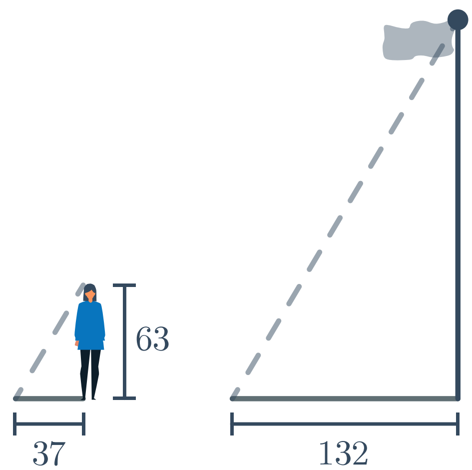
b Hence, determine r, the radius of the circle. Round your answer to one decimal place.



14 A mirror is placed on the ground between an observer and a tree. The measurements are presented in the given figure.
Determine the height of the tree.



15 Tricia wants to find the height of her school's flag pole. During recess, she measures the length of the flag pole's shadow to be 132 in. Her friend then measures her own shadow, which turns out to be 37 in. The triangles formed by casting these shadows are similar.
If Tricia is 63 in tall, solve for the height of the flag pole. Round your answer to the nearest inch.



16 A school building reaching h meters high casts a shadow of 30 m while a 3 m high tree casts a shadow of 6 m.
Solve for the value of h.

17 A 1.1 m high fence casts a shadow of length 1.5 m. At the same time, a lamp post of height 3.9 m casts a shadow of length L meters.
Solve for L, rounding your answer to one decimal place.

18 Council has designed plans for a triangular courtyard in the town square. The drawing shows the courtyard to have dimensions of 4 cm, 6 cm, and 9 cm. The shortest side of the actual courtyard is to be 80 m long.

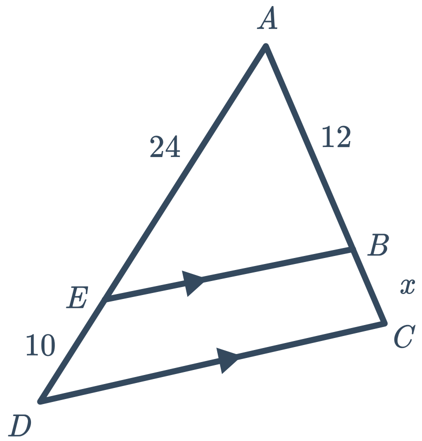
a State the longest side length of the actual courtyard.

b State the middle side length of the actual courtyard in meters.

Additional questions

19 Solve for the variable(s) in each diagram.

a



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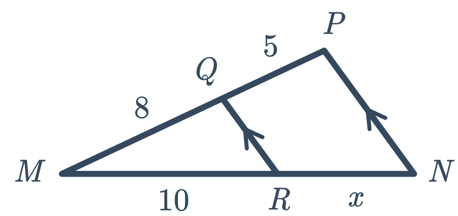

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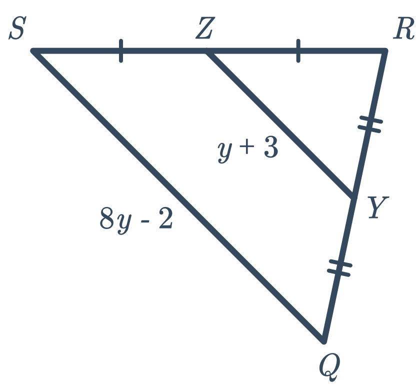

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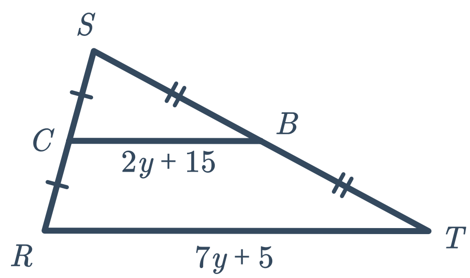

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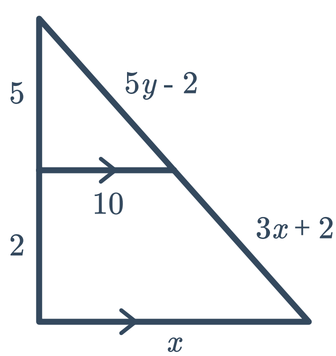

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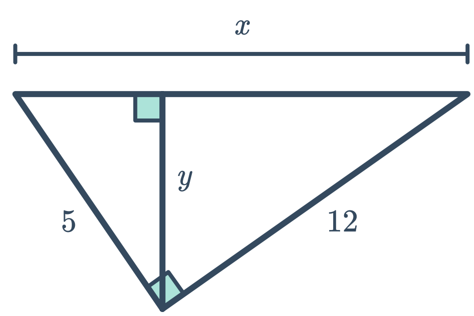

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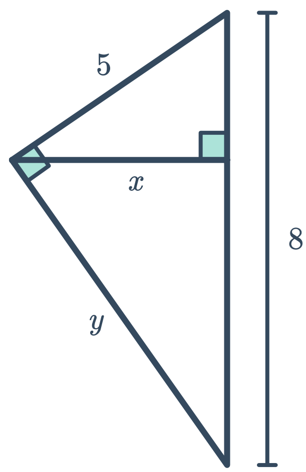
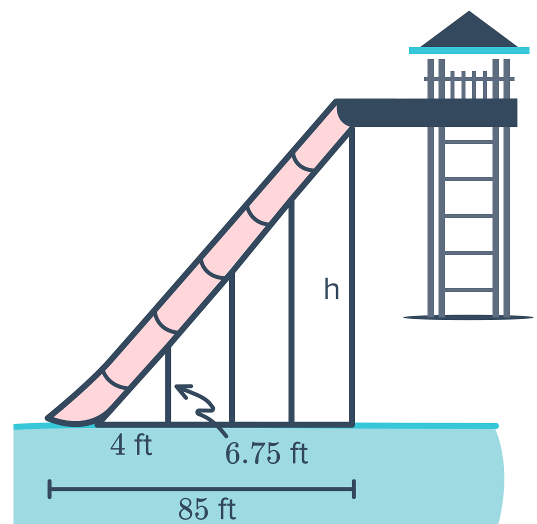


20 The Super Slide at Big Kahuna's Water Park is shown below:
Determine the height of the slide.



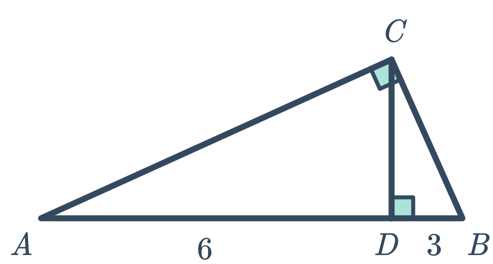
21 Ms. Shin is 5.5 ft tall. She stands near a flag pole that casts a 22 ft long shadow. If Ms. Shin's shadow is 7 ft long, about how tall is the flag pole?

22

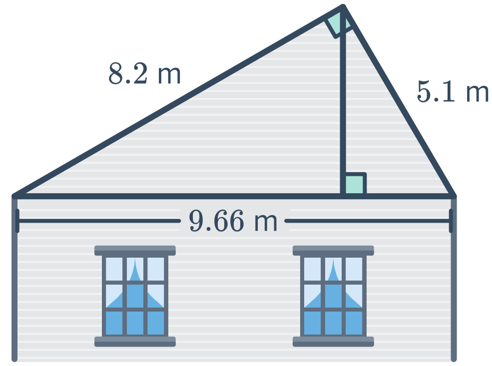
a Solve for $\overline{CD}$

b Solve for $\overline{CB}$

c Solve for $\overline{AC}$

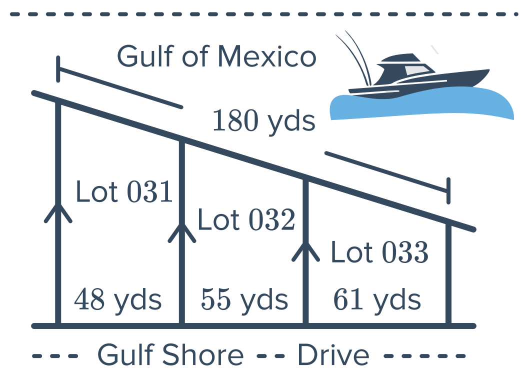


23 Solve for the height of the roof:



24 A developer is selling beachfront lots in Destin. The developer will determine the price of each lot based on how much beachfront access it provides. The greater the beachfront, the higher the price of the lot.

Of the lots shown, which one will have the highest and lowest list prices? Explain.

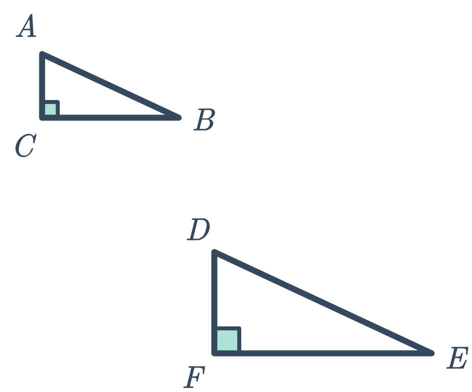


25 Determine whether each of the following is enough to prove $\triangle ABD \sim \triangle CBE$. Justify your answer.

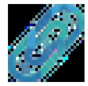
a $\dfrac{AC}{AB} = \dfrac{DF}{DE}$

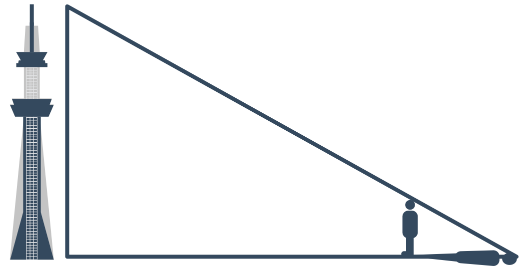
b $\dfrac{CB}{AB} = \dfrac{FD}{DE}$

c $\dfrac{AC}{FE} = \dfrac{AB}{DE}$

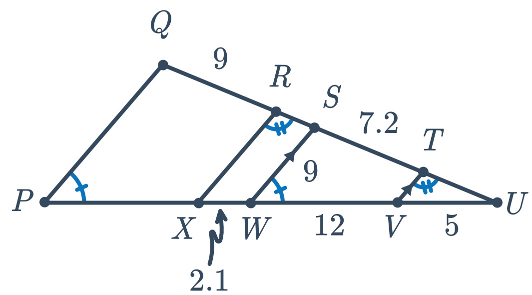


26 Esther visits Tokyo Skytree Tower, which is $2080\ \mathrm{ft}$ tall. If Esther is $5\ \mathrm{ft}\ 3\ \mathrm{in}$ tall, how far away must she stand in order for her shadow to be $\frac{1}{400}$ of her distance to the tower?





27 Solve for all unknown segments:



28 The altitude of a right triangle divides the hypotenuse into two segments. If one of the segments is five times longer than the other, and the length of the altitude is $5\ \mathrm{in}$, what is the length of the hypotenuse?